

Started on	Saturday, 16 May 2026, 10:37 AM
State	Finished
Completed on	Saturday, 16 May 2026, 10:52 AM
Time taken	15 mins 8 secs
Grade	100.00 out of 100.00

Question 1

Correct

Mark 50.00 out of 50.00

Use [LU Decomposition](#) to find L and U matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>

Answer: (penalty regime: 0 %)

Reset answer

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
'''Program to find L and U matrix using LU decomposition.
Developed by: Vemareddygari Pallavi
RegisterNumber: 212225230293
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu
matrix = np.array(eval(input()))
P,L,U=lu(matrix)
print(L)
print(U)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>	✓
✓	[[5, 1, 8], [4, 5, 7], [8, 9, 1]]	<pre>[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]</pre>	<pre>[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]</pre>	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.

Question **2**

Correct

Mark 50.00 out of 50.00

Use [LU Decomposition](#) to solve a matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]

Answer: (penalty regime: 0 %)

Reset answer

Ace editor not ready. Perhaps reload page?

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```
'''Program to solve a matrix using LU decomposition.
Developed by: Vemareddygari Pallavi
RegisterNumber: 212225230293
'''

# To print X matrix (solution to the equations)
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu_factor,lu_solve
matrix=np.array(eval(input()))
constant=np.array(eval(input()))
piv,lu=lu_factor(matrix)
result=lu_solve((piv,lu),constant)
print(result)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]	[0.875 1.125 -0.125]	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.